EXAMPLES OF THE NUMBER SERIS

1.) $\sum \left(\frac{n+1}{2n-1}\right)^n$

$$\lim \sqrt[n]{\left(\frac{n+1}{2n-1}\right)^n} = \lim \frac{n+1}{2n-1} = \frac{1}{2} < 1$$, so the series is convergent

2.) $\sum \frac{1}{n!}$

$$\lim \left|\frac{a_{n+1}}{a_n}\right| = \lim \frac{n!}{n!(n+1)} = 0$$

3.) $\sum \frac{1}{(3n-2)(3n+1)}$

$$\lim \left|\frac{a_{n+1}}{a_n}\right| = \lim \frac{(3n-2)\cancel{(3n+1)}}{\cancel{(3n+1)}(3n+4)} = 1$$, so in conclusive

$$\sum \frac{1}{(3n-2)(3n+1)} < \sum \frac{1}{(3n-2)^2}$$ is convergent majorant

So from the majorant test convergent.

4.) $\sum \frac{1}{n \ln n}$

From the integral test for $f : [2,\infty) \to \mathbb{R}$, $f(n) = a_n$, $f(x) \geq 0$

$$\int_2^\infty \frac{dx}{x \ln x} = \lim_{R \to \infty} \left[\ln|\ln R| - \ln|\ln 2| \right] = +\infty$$ divergent

$$\int \frac{dx}{x \ln x} = \left| \begin{array}{l} t = \ln x \\ dt = \frac{dx}{x} \end{array} \right| = \int \frac{dt}{t} = \ln|\ln x| + C$$